

One-Handed and Off-Balance: Does Proximity Feedback Modality Reduce Falls in a Smartphone Tilt Task?

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[Figure 1 placeholder — game screenshot / system diagram]

Figure 1: The *PathSense* interface. Left: the tightrope balance game displayed on the external monitor. Right: the participant's phone showing proximity feedback and tilt controls. Proximity warning levels (SAFE / NEAR / WARN / DANGER) escalate as the ball approaches the path edge.

Abstract

[Abstract placeholder — to be completed after results are collected.]

CCS Concepts

• **Human-centered computing** → HCI theory, concepts and models; *Haptic devices*.

Keywords

body schema, haptic feedback, motor learning, non-dominant hand, proximity feedback, smartphone, tilt interaction

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1 Introduction

Tilt-based interfaces appear across gaming controllers, accessibility switches, and smartphones, and in many of them the non-dominant hand is the one doing the steering. It holds the device, absorbs corrections, and keeps things stable — tasks that demand real precision from the hand we typically rely on less. The smartphone is the most familiar example, but the pattern is the same wherever tilt control falls to the off-hand: something always feels slightly off. We suggest this reflects a sensory calibration gap, one that graded feedback from the held device may help close. When we actively use a tool, the brain extends its spatial self-model to include the tool's tip [1]. Maravita and Iriki (2004) demonstrated this in macaque monkeys: after five minutes of active rake use, bimodal neurons in the intraparietal cortex expanded their visual receptive fields to span the full length of the tool, as though the

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hand itself had elongated, an effect absent during passive grasping. This mechanism appears to extend to contemporary handheld devices. Lin et al. (2022) showed that smartphones, unlike comparably shaped objects, may specifically integrate into body schema: across five behavioural experiments, participants perceived their forearm as longer after imagined smartphone use and responded faster to smartphone-held hand images than to identically posed hands holding non-smartphone controls [2]. If the held phone is already encoded as a bodily extension, proximity warnings delivered *through* it may update the user's spatial self-model more directly than warnings arriving on a separate channel.

The non-dominant hand is a natural target for this kind of augmentation. Guiard (1987) proposed the kinematic chain model of bimanual action: the non-dominant limb acts as the *frame-setter*, operating at a coarser spatial and temporal scale to establish the postural context within which the dominant hand performs fine-grained movements [3]. The two limbs also differ in how they process proprioceptive information: Adamo and Martin (2009) found asymmetric sensory-motor gains at the wrist, with the non-dominant hand relying more heavily on peripheral proprioceptive feedback while the dominant hand draws on finer cortical resolution [4]. Neural data confirm this asymmetry extends to the neural level. Zabihhosseinian et al. (2020) found that after a novel motor tracing task, the non-dominant hand produced somatosensory evoked potential (SEP) changes consistent with feedback-based impedance control (a 24 % N30 decrease and a 22 % N24 increase), while the dominant hand showed the opposite feedforward pattern [5]. Augmentation should therefore do more work for the limb that already leans on incoming sensory signals.

Whether feedback *modality* shapes that benefit is less clear. Sigrist et al. (2013) reviewed over 120 motor learning studies across visual, auditory, and haptic modalities, concluding that modalities differ substantially in their mechanisms and design requirements [6]. Haptic feedback operates on the same sensorimotor channel as the task itself: the hand that acts also receives the warning, creating a more direct coupling between error signal and corrective movement. Cuppone et al. (2016) provided initial evidence for this: vibrotactile error feedback during proprioceptive training significantly improved wrist joint position-sense acuity within three days, whereas training without it did not, with benefits generalising even to the contralateral limb [7]. Audio warnings, by contrast, are processed on a channel separate from the executing limb; Sigrist et al. (2013) note that while audio can guide continuous movement effectively, it requires deliberate signal-to-correction mapping to avoid misinterpretation [6]. Existing evidence for both modalities comes from lab-grade robots and rehabilitation rigs; whether smartphone-delivered proximity feedback produces comparable effects in a non-dominant hand tilt task remains untested.

We present *PathSense*, a smartphone tightrope tilt task in which participants navigate a ball along a narrow pre-designed path using only their non-dominant hand, receiving graded proximity warnings in one of three modalities (haptic / audio / none). Falls (ball-off-path events) are the primary outcome. We contribute:

- A smartphone-native proximity feedback platform requiring no lab apparatus, enabling modality comparison in a standardised, portable tilt task

- A between-subjects pilot study with a non-dominant-hand constraint across three difficulty levels, providing effect size estimates for a larger planned follow-up
- Preliminary evidence on whether haptic proximity feedback reduces falls more than audio or no feedback when the body schema is engaged through active single-handed phone use

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